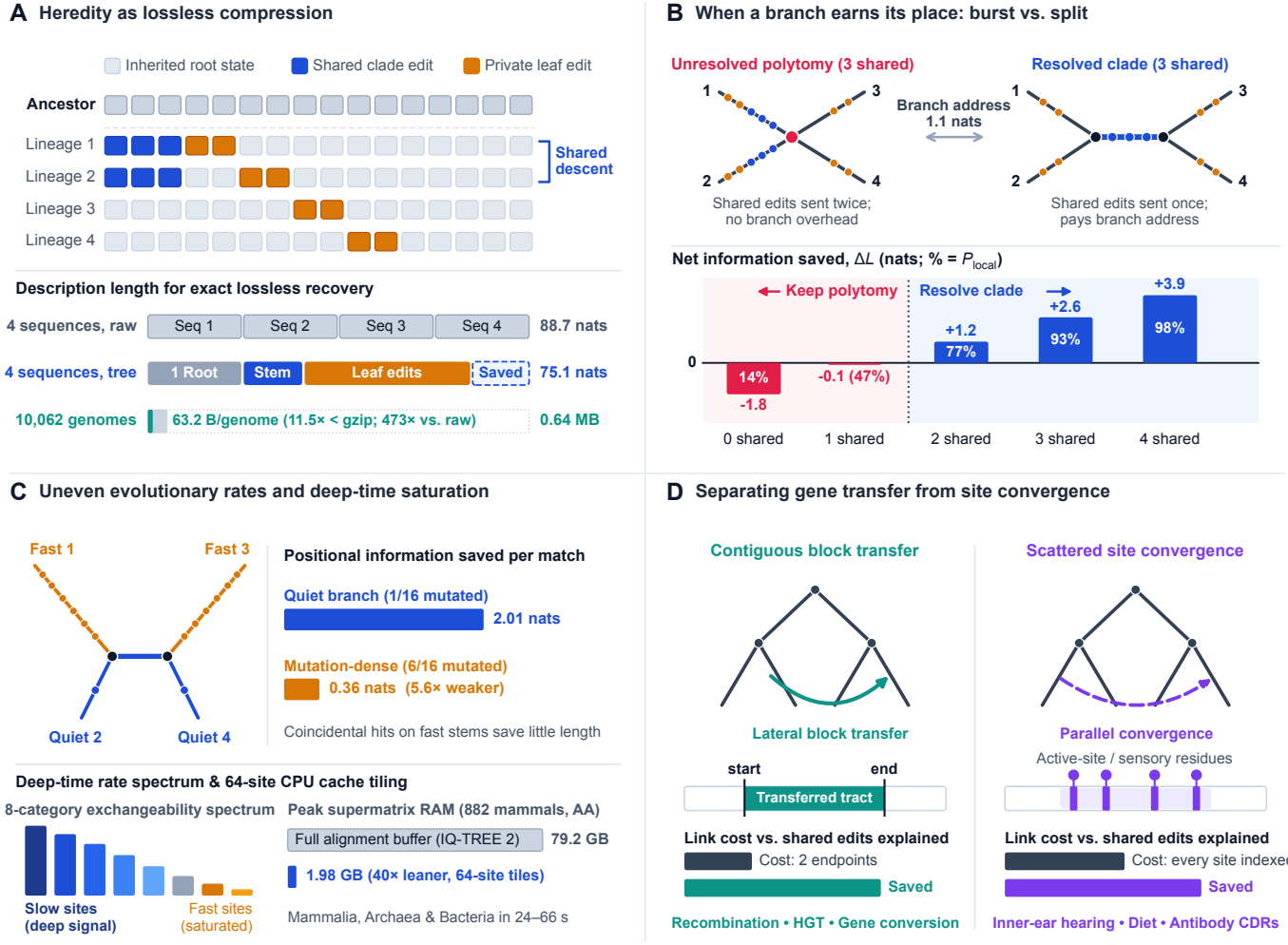

A Heredity as lossless compression
Inherited root state
Shared clade edit
Private leaf edit
Ancestor
Lineage 1
Lineage 2
Lineage 3
Lineage 4
Shared descent
Description length for exact lossless recovery
4 sequences, raw
Seq 1
Seq 2
Seq 3
Seq 4
88.7 nats
4 sequences, tree
1 Root
Stem
Leaf edits
Saved
75.1 nats
10,062 genomes
63.2 B/genome (11.5× < gzip; 473× vs. raw)
0.64 MB
B When a branch earns its place: burst vs. split
Unresolved polytomy (3 shared)
Resolved clade (3 shared)
Branch address
1.1 nats
Shared edits sent twice;
no branch overhead
Shared edits sent once;
pays branch address
Net information saved, ΔL (nats; % = P_local)
Keep polytomy
Resolve clade
+1.2
77%
+2.6
93%
+3.9
98%
14%
-1.8
-0.1 (47%)
0
0 shared
1 shared
2 shared
3 shared
4 shared
C Uneven evolutionary rates and deep-time saturation
Fast 1
Fast 3
Quiet 2
Quiet 4
Positional information saved per match
Quiet branch (1/16 mutated)
2.01 nats
Mutation-dense (6/16 mutated)
0.36 nats (5.6× weaker)
Coincidental hits on fast stems save little length
Deep-time rate spectrum & 64-site CPU cache tiling
8-category exchangeability spectrum
Slow sites
(deep signal)
Fast sites
(saturated)
Peak supermatrix RAM (882 mammals, AA)
Full alignment buffer (IQ-TREE 2)
79.2 GB
1.98 GB (40× leaner, 64-site tiles)
Mammalia, Archaea & Bacteria in 24–66 s
D Separating gene transfer from site convergence
Contiguous block transfer
Lateral block transfer
start
end
Transferred tract
Link cost vs. shared edits explained
Cost: 2 endpoints
Saved
Recombination • HGT • Gene conversion
Scattered site convergence
Parallel convergence
Active-site / sensory residues
Link cost vs. shared edits explained
Cost: every site indexed
Saved
Inner-ear hearing • Diet • Antibody CDRs